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exercise 1:

$$a) \sum_{k=1}^3 k^2 : 1^2 + 2^2 + 3^2 = 1 + 4 + 9 = 14$$

$$b) \sum_{k=1}^3 a_k^2 : a_1^2 + a_2^2 + a_3^2$$

$$c) \sum_{j=1}^3 \sum_{k=1}^3 a_j b_k : a_{11} b_{11} + a_{12} b_{21} + a_{13} b_{31} \\ + a_{21} b_{12} + a_{22} b_{22} + a_{23} b_{32} \\ + a_{31} b_{13} + a_{32} b_{23} + a_{33} b_{33}$$

$$d) \left(\sum_{j=1}^3 a_j \right)^2 = \sum_{j=1}^3 \sum_{k=1}^3 a_j a_k \\ (a_1 + a_2 + a_3)(a_1 + a_2 + a_3) \\ = a_1^2 + a_2^2 + a_3^2 + 2a_1 a_2 + 2a_1 a_3 + 2a_2 a_3$$

Exercise 2

$$v = [1, 2, -5], \quad w = [3, -1, 1]$$

$$a) (v \cdot w) : (1)(3) + (2)(-1) + (-5)(1) = -4$$

$$b) [v \cdot w] : \begin{matrix} & \begin{matrix} x & y & z \end{matrix} \\ \begin{bmatrix} 1 & 2 & -5 \\ 3 & -1 & 1 \end{bmatrix} \end{matrix}$$

$$(2-5)x - (1+15)y + (-1-6)z = -3x - 16y - 7z \\ = [-3, -16, -7]$$

c) Length of v

$$|v| = \sqrt{1^2 + 2^2 + (-5)^2} = \sqrt{1+4+25} \\ = \sqrt{30}$$

$$d) (\delta_1 \cdot v): \delta_1 = [1, 0, 0]$$

$$= [1, 0, 0][1, 2, -5] = 1$$

$$e) [\delta_1 \cdot w]: \begin{bmatrix} \overset{x}{1} & \overset{y}{0} & \overset{z}{0} \\ 3 & -1 & 1 \end{bmatrix} = [0, -1, -1]$$

$$f) \phi_{vw}: \cos \phi = \frac{v \cdot w}{|v||w|}$$

$$|w| = \sqrt{3^2 + (-1)^2 + 1^2} = \sqrt{9+1+1} = \sqrt{11}$$

$$\cos \phi = \frac{-4}{\sqrt{30} \sqrt{11}} = \frac{-4}{\sqrt{330}}$$

$$\phi = \cos^{-1} \left(\frac{-4}{\sqrt{330}} \right)$$

$$g) [r \times v]$$

$$r = (x, y, z)$$

$$r \times v = [x, y, z] \times [1, 2, -5]$$

$$= [-5y - 2z, z + 5x, 2x - y]$$

$$|r \times v| = \sqrt{(-5y - 2z)^2 + (z + 5x)^2 + (2x - y)^2}$$

Exercise 1

a) $[T \cdot v]$

$$\begin{aligned}(T \cdot v)_x &= T_{xx} v_x + T_{xy} v_y + T_{xz} v_z \\ &= 3(5) + 2(3) + (-1)(-2) = 23\end{aligned}$$

$$\begin{aligned}(T \cdot v)_y &= T_{yx} v_x + T_{yy} v_y + T_{yz} v_z \\ &= 2(5) + 2(3) + 1(-2) = 14\end{aligned}$$

$$\begin{aligned}(T \cdot v)_z &= T_{zx} v_x + T_{zy} v_y + T_{zz} v_z \\ &= -1(5) + 1(3) + 4(-2) = -10\end{aligned}$$

$$[T \cdot v] = [23, 14, -10]$$

b) $[v \cdot T]$

$$\begin{aligned}(v \cdot T)_x &= v_x T_{xx} + v_y T_{yx} + v_z T_{zx} \\ &= 5(3) + 3(2) + (-2)(-1) = 23\end{aligned}$$

$$\begin{aligned}(v \cdot T)_y &= v_x T_{xy} + v_y T_{yy} + v_z T_{zy} \\ &= 5(2) + 3(2) + (-2)(1) = 14\end{aligned}$$

$$\begin{aligned}(v \cdot T)_z &= v_x T_{xz} + v_y T_{yz} + v_z T_{zz} \\ &= 5(-1) + 3(1) + (-2)(4) = -10\end{aligned}$$

$$[v \cdot T] = [23, 14, -10]$$

c) $(T: T)$

$$\sum_i \sum_j T_{ij} T_{ij}$$

$$= 3^2 + 2^2 + (-1)^2 + 2^2 + 2^2 + 1^2 + (-1)^2 + 1^2 + 4^2$$

$$T: T = 41$$

d) $(v, [T \cdot v])$

$$v \cdot [T \cdot v] = 5(23) + 3(14) + (-2)(-10)$$

$$= 177$$

e) vv

$$vv = \begin{bmatrix} (5)(5) & (5)(3) & (5)(-2) \\ (3)(5) & (3)(3) & (3)(-2) \\ (-2)(5) & (-2)(3) & (-2)(-2) \end{bmatrix}$$

$$vv = \begin{bmatrix} 25 & 15 & -10 \\ 15 & 9 & -6 \\ -10 & -6 & 4 \end{bmatrix}$$

f) $[T \cdot \delta_1] : \begin{bmatrix} 3 & 2 & -1 \\ 2 & 2 & 1 \\ -1 & 1 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}$

$$T \cdot \delta_1 = [3, 2, -1]$$

Exercise 2

a) $[[\delta_1 \delta_2 \cdot \delta_2] \times \delta_1]$

$$\rightarrow (\delta_1 \delta_2) \cdot \delta_2 = \delta_1 (\delta_2 \cdot \delta_2) = \delta_1 (1) = \delta_1$$

$$\rightarrow [\delta_1 \times \delta_1] = 0$$

b) $(\delta: \delta_1 \delta_2)$

$$\rightarrow \delta: \delta_1 \delta_2 = \delta_{ij} (\delta_1)_i (\delta_2)_j = (\delta_1)_j (\delta_2)_j$$

$$\rightarrow \delta_1 \cdot \delta_2 = 0$$

c) $(\delta: \delta)$

$$\delta: \delta = \delta_{ij} \delta_{ij}$$

$$\rightarrow 1+1+1 = 3$$

d) $(\delta \cdot \delta) : (\delta \cdot \delta)_{ik} = \delta_{ij} \delta_{ik} = \delta_{ik}$

$$= \delta$$

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Exercise 3

a) $\frac{\partial v_x}{\partial x} = 0, \frac{\partial v_x}{\partial y} = b, \frac{\partial v_x}{\partial z} = 0$

$$\nabla \cdot \mathbf{v} = 0 + 0 + 0 = 0$$

$$\nabla \mathbf{v} = \begin{bmatrix} 0 & b & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\nabla \times \mathbf{v} = [0, 0, 0-b] = [0, 0, -b]$$

b) $v_x = bx, v_y = 0, v_z = 0$

$$\frac{\partial v_x}{\partial x} = b, \frac{\partial v_x}{\partial y} = \frac{\partial v_x}{\partial z} = 0$$

$$\nabla \cdot \mathbf{v} = b + 0 + 0 = b$$

$$\nabla \mathbf{v} = \begin{bmatrix} b & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\nabla \cdot \mathbf{v} = [0, 0, 0]$$

$$c) \quad v_x = by, \quad v_y = bx, \quad v_z = 0$$

$$\frac{\partial v_x}{\partial y} = b, \quad \frac{\partial v_x}{\partial x} = 0, \quad \frac{\partial v_y}{\partial x} = b, \quad \frac{\partial v_y}{\partial y} = 0$$

$$\nabla \cdot \mathbf{v} = 0 + 0 + 0 = 0$$

$$\nabla \mathbf{v} = \begin{bmatrix} 0 & b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$(\nabla \times \mathbf{v})_z = \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} = b - b = 0$$

$$\nabla \times \mathbf{v} = [0, 0, 0]$$

$$d) \quad v_x = -by, \quad v_y = bx, \quad v_z = 0$$

$$\frac{\partial v_x}{\partial y} = -b, \quad \frac{\partial v_x}{\partial x} = 0, \quad \frac{\partial v_y}{\partial x} = b, \quad \frac{\partial v_y}{\partial y} = 0$$

$$\nabla \cdot \mathbf{v} = 0 + 0 + 0 = 0$$

$$\nabla \mathbf{v} = \begin{bmatrix} 0 & -b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$(\nabla \times \mathbf{v})_z = \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} = b - (-b) = 2b$$

$$\nabla \times \mathbf{v} = [0, 0, 2b]$$

Exercise 9

$$a) \nabla \frac{1}{r} = -\frac{\mathbf{r}}{r^3}$$

$$r = (x^2 + y^2 + z^2)^{1/2}$$

$$\rightarrow \frac{1}{r} = (x^2 + y^2 + z^2)^{-1/2}$$

$$\frac{\partial}{\partial x} \left(\frac{1}{r} \right) = \frac{\partial}{\partial x} (x^2 + y^2 + z^2)^{-1/2} = -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} (2x) = -\frac{x}{r^3}$$

$$\rightarrow \frac{\partial}{\partial y} \left(\frac{1}{r} \right) = -\frac{y}{r^3}, \quad \frac{\partial}{\partial z} \left(\frac{1}{r} \right) = -\frac{z}{r^3}$$

$$\nabla \left(\frac{1}{r} \right) = i \left(-\frac{x}{r^3} \right) + j \left(-\frac{y}{r^3} \right) + k \left(-\frac{z}{r^3} \right) = -\frac{x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{r^3} = -\frac{\mathbf{r}}{r^3}$$

$$b) \nabla f(r) = \frac{1}{r} \frac{df}{dr} \mathbf{r}$$

$$\frac{\partial f}{\partial x} = \frac{df}{dr} \frac{\partial r}{\partial x}, \quad \frac{\partial f}{\partial y} = \frac{df}{dr} \frac{\partial r}{\partial y}, \quad \frac{\partial f}{\partial z} = \frac{df}{dr} \frac{\partial r}{\partial z}$$

$$\rightarrow \frac{\partial r}{\partial x} :$$

$$r = (x^2 + y^2 + z^2)^{1/2} \rightarrow \frac{\partial r}{\partial x} = \frac{1}{2} (x^2 + y^2 + z^2)^{-1/2} (2x) = \frac{x}{r}$$

$$\frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial r}{\partial z} = \frac{z}{r}$$

$$\frac{\partial f}{\partial x} = \frac{df}{dr} \frac{x}{r}, \quad \frac{\partial f}{\partial y} = \frac{df}{dr} \frac{y}{r}, \quad \frac{\partial f}{\partial z} = \frac{df}{dr} \frac{z}{r}$$

$$\nabla f(r) = i \frac{df}{dr} \frac{x}{r} + j \frac{df}{dr} \frac{y}{r} + k \frac{df}{dr} \frac{z}{r}$$

$$= \frac{df}{dr} \frac{1}{r} (x\mathbf{i} + y\mathbf{j} + z\mathbf{k})$$

$$= \frac{1}{r} \frac{df}{dr} \mathbf{r}$$

c) $\nabla(a \cdot r) = a$ (constant a)

$$a = a_x i + a_y j + a_z k$$

$$a \cdot r = a_x x + a_y y + a_z z$$

take the gradient

$$\nabla(a \cdot r) = i \frac{\partial}{\partial x} (a_x x + a_y y + a_z z) + j \frac{\partial}{\partial y} (a_x x + a_y y + a_z z) + k \frac{\partial}{\partial z} (a_x x + a_y y + a_z z)$$

$$\frac{\partial}{\partial x} (a_x x + a_y y + a_z z) = a_x, \quad \frac{\partial}{\partial y} (a_x x + a_y y + a_z z) = a_y, \quad \frac{\partial}{\partial z} (a_x x + a_y y + a_z z) = a_z$$

$$\nabla(a \cdot r) = a_x i + a_y j + a_z k = a$$

Exercise 10

a) $\frac{\partial}{\partial t} p v = -[\nabla \cdot p v v] - \nabla p - [\nabla \cdot \tau] + p g$

$$(\nabla \cdot (p v v))_i = \frac{\partial (p v_i v_x)}{\partial x} + \frac{\partial (p v_i v_y)}{\partial y} + \frac{\partial (p v_i v_z)}{\partial z}$$

$$(\nabla p)_x = \frac{\partial p}{\partial x} \dots \dots \dots$$

$$(\nabla \cdot \tau)_x = \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z}, \dots \dots \dots$$

$$\frac{\partial (p v_x)}{\partial t} = - \left[\frac{\partial (p v_x v_x)}{\partial x} + \frac{\partial (p v_x v_y)}{\partial y} + \frac{\partial (p v_x v_z)}{\partial z} \right] - \frac{\partial p}{\partial x} - \left[\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} \right] + p g_x$$

$$\frac{\partial (p v_y)}{\partial t} = - \left[\frac{\partial (p v_y v_x)}{\partial x} + \frac{\partial (p v_y v_y)}{\partial y} + \frac{\partial (p v_y v_z)}{\partial z} \right] - \frac{\partial p}{\partial y} - \left[\frac{\partial \tau_{yx}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} \right] + p g_y$$

$$\frac{\partial (p v_z)}{\partial t} = - \left[\frac{\partial (p v_z v_x)}{\partial x} + \frac{\partial (p v_z v_y)}{\partial y} + \frac{\partial (p v_z v_z)}{\partial z} \right] - \frac{\partial p}{\partial z} - \left[\frac{\partial \tau_{zx}}{\partial x} + \frac{\partial \tau_{zy}}{\partial y} + \frac{\partial \tau_{zz}}{\partial z} \right] + p g_z$$

$$b) \quad T = -\mu \left[\nabla v + (\nabla v)^T - \frac{2}{3} (\nabla \cdot v) \delta \right]$$

$$\rightarrow \nabla \cdot v = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$$

$$T_{ij} = -\mu \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} - \frac{2}{3} (\nabla \cdot v) \delta_{ij} \right)$$

$$\cdot T_{xx} = -\mu \left(2 \frac{\partial v_x}{\partial x} - \frac{2}{3} \nabla \cdot v \right)$$

$$\cdot T_{yy} = -\mu \left(2 \frac{\partial v_y}{\partial y} - \frac{2}{3} \nabla \cdot v \right)$$

$$\cdot T_{zz} = -\mu \left(2 \frac{\partial v_z}{\partial z} - \frac{2}{3} \nabla \cdot v \right)$$

$$\delta_{ij} = 0 \text{ for } i \neq j$$

$$T_{xy} = T_{yx} = -\mu \left(\frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x} \right)$$

$$T_{xz} = T_{zx} = -\mu \left(\frac{\partial v_x}{\partial z} + \frac{\partial v_z}{\partial x} \right)$$

$$T_{yz} = T_{zy} = -\mu \left(\frac{\partial v_y}{\partial z} + \frac{\partial v_z}{\partial y} \right)$$

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$$v = \delta_1 x_1 + \delta_2 x_2 + \delta_3 x_3$$

$$v_1 = x_1, \quad v_2 = x_3, \quad v_3 = x_2$$

$$\int_V (\nabla \cdot v) dV = \oint_S v \cdot n ds$$

left hand side

$$\int_V (\nabla \cdot v) dV$$

$$\nabla \cdot v = \frac{\partial v_1}{\partial x_1} + \frac{\partial v_2}{\partial x_2} + \frac{\partial v_3}{\partial x_3}$$

$$= \frac{\partial x_1}{\partial x_1} + \frac{\partial x_3}{\partial x_2} + \frac{\partial x_2}{\partial x_3}$$

$$= 1 + 0 + 0 = 1$$

$$\int_V 1 dV = \int_0^1 \int_0^2 \int_0^4 dx_3 dx_2 dx_1$$

$$= 1 \times 2 \times 4 = 8$$

Right hand side

$$\oint_S v \cdot n ds$$

faces normal to x_1

$$x_1 = 0, \quad n = -\delta_1$$

$$v \cdot n = -v_1 = -x_1 = 0 \rightarrow \text{flux} = 0$$

$$x_1 = 1, \quad n = +\delta_1$$

$$v \cdot n = v_1 = 1$$

$$\text{Area} = 2 \times 4 = 8$$

$$\text{flux} = 1 \times 8 = 8$$

faces normal to x_2

$$x_2 = 0, \quad n = -\delta_3$$

$$v \cdot n = -v_3 = -x_2 = 0 \rightarrow 0$$

$$x_2 = 2, \quad n = +\delta_3$$

$$v \cdot n = v_3 = 2$$

$$\text{Area} = 1 \times 4 = 4$$

$$\text{flux} = 2 \times 4 = 8$$

faces normal to x_3

$$x_3 = 0, \quad n = -\delta_2$$

$$v \cdot n = -v_2 = -x_3 = 0 \rightarrow 0$$

$$x_3 = 4, \quad n = +\delta_2$$

$$v \cdot n = v_2 = 4$$

$$\text{Area} = 1 \times 2 = 2$$

$$\text{flux} = 4 \times 2 = 8$$

$$\int_V (\nabla \cdot v) dV = \oint_S v \cdot n dS = 8$$

Page 839. Exercise 1

$$\frac{\partial \delta r}{\partial \theta} = \delta \theta, \quad \frac{\partial \delta \theta}{\partial \theta} = -\delta r, \quad \frac{\partial \delta z}{\partial \theta} = 0 \quad \text{these are spatial derivatives, not time derivatives}$$

to become a time derivative

$$\dot{\delta r} = \dot{\theta} \delta \theta, \quad \dot{\delta \theta} = -\dot{\theta} \delta r, \quad \dot{\delta z} = 0$$

Position vector

$$\mathbf{r} = r \delta r + z \delta z$$

Velocity

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = \dot{r} \delta r + r \dot{\delta r} + \dot{z} \delta z$$

$$\rightarrow \delta r = \dot{\theta} \delta \theta$$

$$\mathbf{v} = \delta r \dot{r} + \delta \theta r \dot{\theta} + \delta z \dot{z}$$

Exercise 3

a) $(\nabla \cdot \mathbf{pV})$, (Table A.7-2, (A))

$$\nabla \cdot (\mathbf{pV}) = \frac{1}{r} \frac{\partial}{\partial r} (r p v_r) + \frac{1}{r} \frac{\partial (p v_\theta)}{\partial \theta} + \frac{\partial (p v_z)}{\partial z}$$

b) $[\nabla \cdot (\mathbf{pV})]_r$, (Table A.7-2 (P-R))

$$[\nabla \cdot (\mathbf{pV})]_r = \rho \left[v_r \frac{\partial v_r}{\partial r} + v_\theta \left(\frac{1}{r} \frac{\partial v_r}{\partial \theta} - \frac{v_\theta}{r} \right) + v_z \frac{\partial v_r}{\partial z} \right]$$

$$[\nabla \cdot (\mathbf{pV})]_\theta = \rho \left[v_r \frac{\partial v_\theta}{\partial r} + v_\theta \left(\frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right) + v_z \frac{\partial v_\theta}{\partial z} \right]$$

$$[\nabla \cdot (\mathbf{pV})]_z = \rho \left[v_r \frac{\partial v_z}{\partial r} + v_\theta \frac{1}{r} \frac{\partial v_z}{\partial \theta} + v_z \frac{\partial v_z}{\partial z} \right]$$

c) $[\nabla \cdot \rho \delta]_\theta$, (Table A.7-2(A))

$$\nabla \cdot (\rho \delta_\theta) = \frac{1}{r} \frac{\partial \rho}{\partial \theta}$$

d) $(\nabla \cdot [\tau \cdot v])$, (Table A.7-2 (J-L))

$$\nabla \cdot (\tau \cdot v) = \frac{1}{r} \frac{\partial}{\partial r} (r \tau_{rr} v_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (\tau_{r\theta} v_\theta) + \frac{\partial}{\partial z} (\tau_{rz} v_z) - \frac{\tau_{\theta\theta} v_r}{r}$$

e) $[v \cdot \nabla v]_\theta$ (a)

$$[v \cdot \nabla v]_\theta = v_r \frac{\partial v_\theta}{\partial r} + v_\theta \left(\frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right) + v_z \frac{\partial v_\theta}{\partial z}$$

f) $\nabla v + (\nabla v)^T$ (S-AA)

$$(\nabla v + (\nabla v)^T)_{ij} = \frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}$$

$$\cdot (\nabla v + (\nabla v)^T)_{rr} = 2 \frac{\partial v_r}{\partial r}$$

$$\cdot (\nabla v + (\nabla v)^T)_{r\theta} = \frac{1}{r} \frac{\partial v_r}{\partial \theta} + \frac{\partial v_\theta}{\partial r} - \frac{v_\theta}{r}$$

$$\cdot (\nabla v + (\nabla v)^T)_{\theta\theta} = 2 \left(\frac{1}{r} \frac{\partial v_\theta}{\partial \theta} + \frac{v_r}{r} \right)$$

$$\cdot (\nabla v + (\nabla v)^T)_{rz} = \frac{\partial v_r}{\partial z} + \frac{\partial v_z}{\partial r}$$

Using the unit vectors of a Cartesian coordinate system, prove the following useful identities:

- a- $\operatorname{div}(\rho \langle v \rangle) = \langle \nabla \rho | v \rangle + \rho \operatorname{div} \langle v \rangle$
- b- $\operatorname{div} \langle v \rangle \langle w \rangle = |\nabla v| w + \langle \operatorname{div} w \rangle v$
- c- $\operatorname{div} \langle v | \tau \rangle = |\tau| |\nabla v| + \langle v | \operatorname{div} \tau \rangle$

Hint: use the product rule for differentiation: $\frac{\partial(fg)}{\partial x} = g \frac{\partial f}{\partial x} + f \frac{\partial g}{\partial x}$

$$a) \quad \nabla \cdot (\rho v) = (\nabla \rho) \cdot v + \rho \nabla \cdot v$$

$$\rho v = \rho v_i e_i$$

$$\rightarrow \nabla \cdot (\rho v) = \frac{\partial}{\partial x_i} (\rho v_i)$$

$$= v_i \frac{\partial \rho}{\partial x_i} + \rho \frac{\partial v_i}{\partial x_i}$$

$$v_i \frac{\partial \rho}{\partial x_i} = (\nabla \rho) \cdot v, \quad \frac{\partial v_i}{\partial x_i} = \nabla \cdot v$$

$$\nabla \cdot (\rho v) = (\nabla \rho) \cdot v + \rho \nabla \cdot v$$

$$b) \quad \nabla \cdot (v w) = (\nabla v) \cdot w + (\nabla \cdot w) v$$

$$(v w)_{ij} = v_i w_j$$

$$\rightarrow w_j \frac{\partial v_i}{\partial x_j} + v_i \frac{\partial w_j}{\partial x_j}$$

$$\nabla \cdot (v w) = (\nabla v) \cdot w + (\nabla \cdot w) v$$

$$c) \quad \nabla \cdot (vT) = T : \nabla v + v (\nabla \cdot T)$$

$$vT : (vT)_{ij} = v_i T_{ij}$$

$$\rightarrow T_{ij} \frac{\partial v_i}{\partial x_j} + v_i \frac{\partial T_{ij}}{\partial x_j}$$

$$\nabla \cdot (vT) = T : \nabla v + v (\nabla \cdot T)$$